

Prerequisite Test

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Linear Algebra

1. Absolute error = $2.2 - 2 = \underline{0.2}$

Relative error = $.2/2 = .1$

$$2. a. y = Ab = \begin{bmatrix} -3 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ -3 \end{bmatrix}$$

$$6. Ax = \begin{bmatrix} -3 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = b$$

$$\begin{bmatrix} -3 & 1 & 0 & | & 1 \\ 1 & -2 & 1 & | & 1 \\ 0 & 1 & -2 & | & 2 \end{bmatrix} + 3R_2$$

$$\begin{bmatrix} 0 & -5 & 3 & 4 \end{bmatrix} + 5R_2$$

$$\begin{bmatrix} 0 & 0 & -7 & | & 15 \\ 1 & -2 & 1 & | & 1 \\ 0 & 1 & -2 & | & 2 \end{bmatrix} \cdot 7$$

$$\left[\begin{array}{ccc|c} 0 & 0 & 1 & -2 \\ 1 & -2 & 1 & 1 \\ 0 & 1 & -2 & 2 \end{array} \right] + 2R_3$$

$$\rightarrow \begin{bmatrix} 0 & 0 & 1 & -2 \\ 1 & -2 & 1 & 1 \\ 0 & 1 & 0 & -2 \end{bmatrix} \quad +2R_3 - R_1$$

$$\begin{bmatrix} 0 & 0 & 1 & | & -2 \\ 1 & 0 & 0 & | & -1 \\ 0 & 1 & 0 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & | & -1 \\ 0 & 1 & 0 & | & -2 \\ 0 & 0 & 1 & | & -2 \end{bmatrix}$$

$$X = \begin{bmatrix} -1 \\ -2 \\ -2 \end{bmatrix}$$

I solved by setting up an augmented matrix and performing gaussian elimination.

3. a. n and s must be equal

b. There's no condition on the dimensions of A because A^T is $n \times m$ and A is $m \times n$ and $m = m$ is always true. The dimensions of AA^T are $n \times n$.

3. c. $\mathbb{R}^{m \times n} \cdot \mathbb{R}^{n \times t} = \mathbb{R}^{s \times t}$

$x \in \mathbb{R}^{n \times t}$ (m must equal s for this to be possible)
 d. $\mathbb{R}^{t \times s} \cdot \mathbb{R}^{n \times m} = \mathbb{R}^{t \times m}$ or $B^T A^T = (AB)^T \in (\mathbb{R}^{m \times t})^T = (\mathbb{R}^{t \times m})$
 $B^T A^T \in \mathbb{R}^{t \times m}$

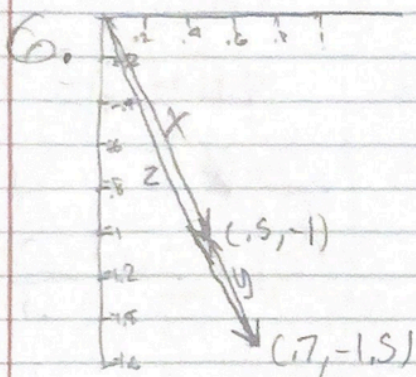
4. 0

5. a. Infinite

b. 0 (assuming system isn't consistent)

c. 1

d. 0 (assuming system isn't consistent)



Calculus

1. $\lim_{n \rightarrow \infty} a_n = a^*$

Where g remains bounded if $h \geq 0$

2. $e(h) = O(h^2)$ if $e(h) = g(h) \cdot h^2$

This means that the truncation error can be approximated to the same bounded function with error of order 2.

$$3. \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$4. \frac{d}{dx} \left(\frac{1}{x} \right) = \frac{d}{dx} (x^{-1}) = -x^{-2} = -\frac{1}{x^2}$$

$$-\frac{1}{(.5)^2} = \boxed{-4} = \frac{d}{dx} \left(\frac{1}{x} \right) \text{ at } x = .5$$

This means that the rate of change of the function $y = 1/x$ can be modeled by the function $y = -1/x^2$, and the rate of change at $x = .5$ of $y = 1/x$ is -4 .

$$5. \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n$$

$$6. \int_1^2 \ln x \, dx$$

$$\int u \, dv = uv - \int v \, du$$

$$u = \ln(x) \quad du = \frac{1}{x} \, dx$$

$$dv = \frac{1}{x} \, dx \quad v = x$$

$$\int_1^2 \ln(x) \, dx = x \ln(x) - \int_1^2 \frac{x}{x} \, dx$$

$$= x(\ln(x) - 1) \Big|_1^2$$

$$= 2(\ln(2) - 1) + 1$$

$$= .386$$

This is the area under the function $y = \ln(x)$ from $x = 1$ to $x = 2$

$$7. \frac{du}{dt} = au$$

$$\int \frac{1}{u} \, du = \int a \, dt$$

$$\ln|u| = at + c$$

$$u(t) = Ce^{at}$$

$$\text{If } a > 0, \lim_{t \rightarrow \infty} u(t) = \infty$$

$$\text{If } a < 0, \lim_{t \rightarrow \infty} u(t) = 0$$

$$8. \frac{d}{dx}(-x^2) = -2x = f'(x)$$

$$-2x = 0 \text{ at } x = 0$$

$$f(x) = -x^2$$

relative max at $x = 0$

Check relative extrema & endpoints:

$$f(-1) = -1$$

$$f(0) = 0$$

$$f(1) = -1$$

$$\text{min: } y = -1 \text{ at } x = -1$$

$$\text{max: } y = 0 \text{ at } x = 0$$

Coding

Functions and Vectors

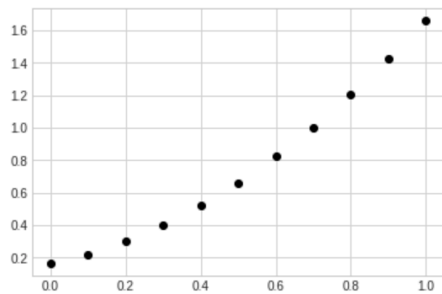
So, I created a vector with 11 equidistant points that cover $[0, 1]$, made a new vector that represents $y = x^2 - .5x + .1625$ for all the points in the first vector, plotted the two vectors in a scatterplot, and used a for loop to calculate the sum of the y values, the max y value, and the square root of the sum of the squares of the y values.

Data

2. The distance between the points appears to first decrease then increase at a linearly increasing rate. This could be a quadratic function.

```
[9]: %matplotlib inline
import matplotlib.pyplot as plt
plt.style.use('seaborn-whitegrid')
import numpy as np
import math
myList = [0.0,0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9,1.0]
myVector = np.array(myList)
myNewVector = (myVector * myVector) + (myVector * 0.5) + 0.1625
plt.plot(myVector, myNewVector, 'o', color='black');
maxY = -99999.0
sumY = 0.0
sumSquares = 0.0
for x in range(11):
    sumY += myNewVector[x]
    sumSquares += (myNewVector[x]) ** 2
    if myNewVector[x] > maxY :
        maxY = myNewVector[x]
sqrtSumSquares = math.sqrt(sumSquares)
print("Σjyj = %f" %(sumY))
print("maxjyj = %f" %(maxY))
print("(Σjy2j)1/2 = %f" %(sqrtSumSquares))
```

Σjyj = 8.387500
maxjyj = 1.662500
(Σjy2j)1/2 = 2.992703



```
[2]: %matplotlib inline
import matplotlib.pyplot as plt
plt.style.use('seaborn-whitegrid')
import numpy as np
#data from data.txt put into list1 and list2
list1 = [0,0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9,1]
list2 = [0.161258556517837,0.137396976077855,0.116590344898005,0.116671924134296,0.129214971336081,0.15042513077315,0.229
vector1 = np.array(list1)
vector2 = np.array(list2)
plt.plot(vector1, vector2, 'o', color='black');
```

